

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester II Examination, 2013-14 (Old Course)

Paper No. CHB-201: Inorganic & Physical Chemistry-I

Time: 3 Hours

Maximum Marks: 70

Instructions: Answer five questions in total, including Question 1 from each section which is compulsory. Use separate answer books for Section A and Section B.

SECTION A

(Inorganic Chemistry-I – Marks: 35)

Question 1. Answer all parts of the following:

[5 × 2 = 10]

- (a) What is electrode potential? Explain the term with the help of a Born-Haber type cycle.
- (b) What happens when alkali metal tetrahydroborates are treated with water?
- (c) Draw the structures of Cryptand-2,2,2 and Benzo-18-crown-6.
- (d) Draw the structure of basic beryllium acetate.
- (e) Draw and explain the structure of XeF_5^- .

Question 2.

- (a) Derive the Born-Landé equation for the lattice energy of an ionic crystal. Explain all the terms involved. [6]
- (b) Discuss the solvation energy of ions. What are the factors that influence the solvation of ions? [6]

Question 3.

- (a) What are silicates? How are they classified? Give the structures of orthosilicates and chain silicates. [6]
- (b) Write short notes on: [6]
 - (i) Intercalation compounds of graphite
 - (ii) Nitrogen activation reactions

Question 4.

- (a) Discuss the bonding, structure, and applications of polymeric sulfur nitride $((\text{SN})_x)$
[6]
- (b) Draw structures, predict hybridizations, and explain the shapes of the following: (i) XeF_6 , (ii) XeOF_4 , (iii) XeO_4 . [6]

SECTION B

(Physical Chemistry-I – Marks: 35)

Question 5. Answer all parts of the following: [5 × 2 = 10]

- (a) State the law of corresponding states and write its reduced equation.
- (b) Explain why the surface tension of a liquid decreases with an increase in temperature.
- (c) Explain the terms: average velocity, most probable velocity, and root-mean-square velocity.
- (d) State the first law of thermodynamics. Define heat capacity at constant volume (C_v) and constant pressure (C_p).
- (e) Under what conditions does a real gas behave like an ideal gas?

Question 6.

- (a) Define the surface tension of a liquid. Discuss the stalagmometer method for the determination of surface tension. What is the influence of temperature on surface tension? [8]
- (b) Water rises to a height of 7.4 cm in a glass capillary at 20°C. Calculate the radius of the capillary. (Surface tension of water = 72.8 dynes/cm, density = 1 g/cm³, $g = 981 \text{ cm/s}^2$). [4]

Question 7.

- (a) Using the Maxwell-Boltzmann distribution of molecular speeds, derive expressions for average, most probable, and root-mean-square velocities of gas molecules. [6]
- (b) Discuss the kinetic theory of gases. Show how the kinetic gas equation can be used to derive Dalton's law of partial pressures. [6]

Question 8.

- (a) What is the Joule-Thomson effect? Show that the Joule-Thomson expansion of a real gas is an isenthalpic process. [6]
- (b) Calculate the work done in the isothermal reversible expansion of 2 moles of an ideal gas from 2 L to 20 L at 300 K. [6]

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